Design of Smart Device for Foot of Diabetic Patient in Malaysia

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2021 International Congress of Advanced Technology and Engineering (ICOTEN) | 978-1-6654-1224-7/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICOTEN52080.2021.9493450

***Abstract*— A diabetic smart device is a smart insole that is exclusively developed for diabetic patients in Malaysia. This device focuses on displaying the foot plantar pressure of its user and the temperature of the foot through smartphone. Diabetic foot wounds are caused by uncontrolled activity and repetitive stress to the foot that can cause some damage to the skin. Hence, this device aims to limit its user in certain amount of pressure while they are doing other activities. User can download Blynk application in their smartphone and connect it with the device via Bluetooth. The system provides the wearer with real-time warning message that allows them to take actions immediately in keeping their feet healthy as well as preventing from diabetic foot ulceration. In this work, the pressure with temperature data from foot plantar has been displayed via smartphone and the feedback system that can warn the user has been completed. In conclusion, the smart insole system has gone through several testing and it needs some improvement in ensuring good stability and accuracy of the hardware used in future works.**

Keywords—diabetic smart device, pressure, temperature, diabetic foot ulceration

I. Introduction

Diabetes is a serious disease that happens when the pancreas in our body does not produce enough insulin where insulin is a hormone that regulates blood sugar. The total cases of diabetes among adults in Malaysia are more than 3.5 million this year and the number is increasing year by year [1]. In fact, there are approximately 15 percent of patients with diabetes are suffering the diabetic foot ulcer that occurs on foot plantar [2].

Over the past few years, the need for progressive methods in reducing the foot ulceration among the diabetic patients as well as for preventing the ulceration has led to the development of new design of shoes and innovation foot sensor [3]. Diabetic foot ulceration is very synonyms to pressure ulcers on the foot occurring in people with diabetes. The pressure and temperature sensing techniques are playing the important roles in the medical innovation technologies [4, 5].

This paper presents the design and development of a smart insole device. There are two objectives are set to be achieved which to display the pressure and temperature reading for foot diabetic patient through smartphone and to activate feedback system senses via smartphone that can warn the user. To achieve the objectives, several tests have been made and the results have been recorded and compared in this paper.

II. Literature Review

diabetic foot ulcer (DFU) is defined as a full thickness wound below the ankle in a diabetic patient. There are three main causes of the ulceration: Neuropathy, Ischaemia and Shear [6]. Patients with neuropathy must be very careful when fitting footwear for example shoes should not be too tight or too loose because inappropriate footwear is a major cause of ulceration [7]. High plantar pressure in diabetic patients have a strong relationship with the plantar ulceration happened especially when the patients have neuropathy problem. Therefore, the foot pressure measurement can be used to predict the foot ulceration happen in the future [8].

Figure 1 illustrates the dynamic plantar pressure footprints generated by each patient that were categorized into 10 masks which are heel (M01), midfoot (M02), 1st metatarsal (M03), 2nd metatarsal (M04), 3rd metatarsal (M05), 4th metatarsal (M06), 5th metatarsal (M07), hallux (M08), 2nd toe (M09) and toes 3-5 (M10) [9]. Each of the colour in Figure 1 below shows their maximum or peak pressure value on the foot plantar area.

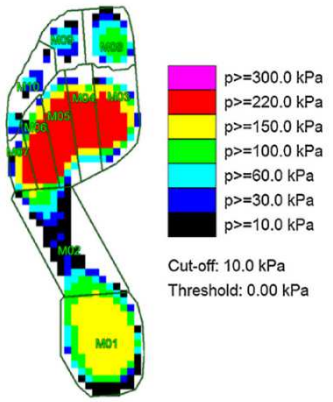

Fig. 1. Example of a mean peak pressure picture showing the mask regions [9].

Footwear is beneficial in reducing neuropathic complications among geriatric patients with diabetic foot problem. Diabetic shoe shows more effective than regular footwear in reducing the pain due to neuropathy. It also can help the patients to reduce the risk of diabetic ulcers and avoid foot amputation [10]. Ulcer area off-loading is extremely important for the healing of plantar ulcers [11]. the idea of offloading the foot was introduced as a treatment strategy after numerous surgical and non-operational procedures. There have been many offloading tools were designed to reduce the pressure on the foot area [12].

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Besides that, the probability for the patients to experience the foot ulceration is higher if they have high temperatures under the foot with diminished loss of sensation [13]. Temperature monitoring has been highly suggested as a clinical tool to monitor diabetic patient especially in foot area [14]. Temperature foot monitoring has been proven as an effective way to provide an accurate guidance that will allow patients to adjust their activity and secure their feet before ulceration occur [15].

There are many studies that have been done based on the smart insole device concept that employs many sensing methods to prevent the ulceration among diabetic patients. Most of the research studies use either Internet of things (IoT) or Bluetooth communication system in their proposed device. In a paper by Malvade et al [16], Arduino UNO is being used as the microcontroller to get the pressure data from foot plantar pressure through FSR402 pressure sensor. After that, it will send the data to the smartphone via Bluetooth and then save the data to the cloud through IoT.

III. Propose System Device

A. Placement of Sensors and Insole Design

4 pressure sensors are considered in collecting the pressure data and were placed at the spots which are very important such as at the big toe, two sensors at metatarsal head and one more at heel. Based on paper by Pataky et al [17], there are 5 spots on foot plantar that have high pressure which are big toe, first metatarsal, third metatarsal, fifth metatarsal and heel. Figure 2 below shows the placement of 4 pressure sensors and one temperature sensor LM35 on the insole.

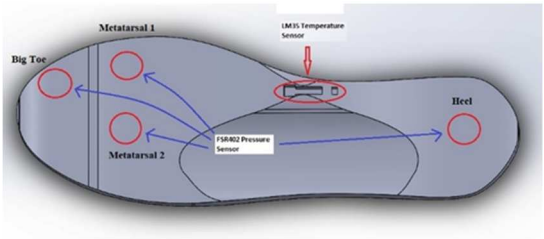

Fig. 2. Top view insole design

LM35 has been put at that location because that area has the thickest insole space that will not affect the foot condition once the user step on it. This is intended to take a reading from the foot as the temperature can also be a sign before the foot ulceration occurs.

B. System Operation

The system is designed to measure the pressure level and temperature value of the foot diabetic patients. Initially, the microcontroller which is the Arduino UNO will take the data reading from the pressure sensors and temperature sensor. Then, the data will be transmitted using Bluetooth module HC-05 and received by Bluetooth receiver from the smartphone. Blynk app in the smartphone will be used to display the real-time reading of the sensors. The system design can be seen on Figure 3 below.

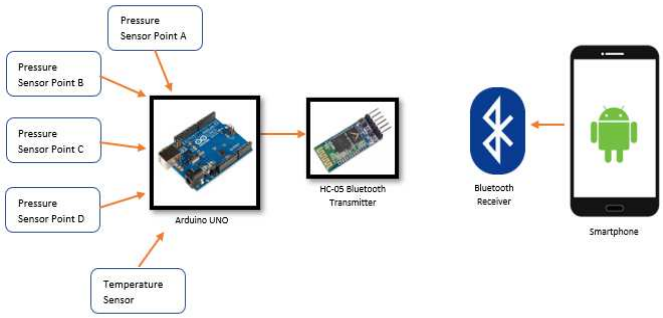

Fig. 3. Data communication system

For the software implementation, Arduino IDE was used to design the coding for the sensors to work well in detecting the pressure and temperature values of the foot. Then, Blynk App is used as the interface for monitoring system on the smartphone that will display the data taken from the sensors.

C. Circuit Connection

Arduino UNO is used as the main control (processing) unit and it is still suitable to be used in this system because this circuit only use six sensors that are connected to six input ports of the microcontroller. The circuit will be connected to the Bluetooth module by connecting the module to the transmitter and receiver ports of the microcontroller. Figure 4 shows the circuit connection for this device.

As can be seen in the circuit above, voltage divider method is used for the FSR sensor to convert the resistance value from the sensor to output voltage. Besides that, the working principle of the FSR is that when a load force is push on it, the resistance of the FSR will decrease. So, 10K ohm resistor is used in this circuit to make sure that when full load force is push on the sensor, the total resistance of the FSR decreases from the maximum resistance to the minimum resistance which is 10K ohm.

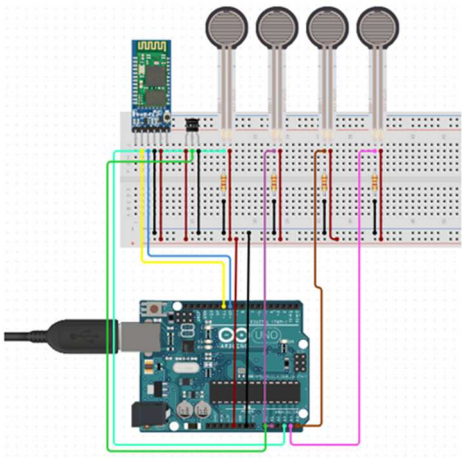

Fig. 4. Circuit connection

IV. Measurement Setup

A. Hardware Setup

The circuit and all components for this system except the sensors were connected on a breadboard inside a blue box as shown in Figure 5 below. 9V battery was used to supply the Arduino UNO while the Bluetooth module HC-05 acts as the data transmitter.

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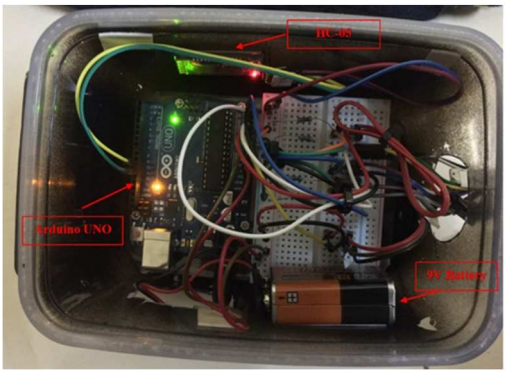

Fig. 5. Components inside blue box

Besides, the insole will be covered to comfort the user's foot as well as to cover the surface of the FSR sensor. Then, the smart insole will replace the normal insole inside the shoes. After that, user can wear the device and tighten the blue box on the calf using a belt provided. Figure 6 illustrates the smart insole set-up and how to wear the device.

(a) (b)

Fig. 6. Hardware set-up for the system (a) The look of the device (b) Way of wearing the device

B. Blynk Application Setup

Blynk is a platform that allows user to create applications and then use it to control Arduino board that are connected to a PC with internet access, from anywhere around the world with a smartphone. The connection can also be established by Bluetooth between smartphone and Arduino board. Blynk is a digital dashboard where it can build a graphic interface for the user by simply drag and drop the widgets.

First, the Blynk apps must be downloaded and installed from Google plays store or app store for android or iOS. Besides that, user needs to install Blynk library at library folder of Arduino IDE to interface it with Arduino UNO. After finished the installation process, the apps can be open on the smartphone and click the "create new account" to login for a new project. For this project, the hardware selected is Arduino UNO and the communication type is Bluetooth. To build a project, the auth-token needs to be obtained first via email account. Once the user obtaining the auth-token, user can proceed to design the Blynk apps. Once confirmed, the user can setup the button widget where each of the widgets has its own settings.

Figure 7 shows the data reading from the pressure and temperature sensors displayed in Blynk application. There are one temperature display and four pressure display at different area of foot. For the temperature reading, some calculation is used in the coding to get the actual temperature at the area. For the pressure reading, there are some limitation for this FSR sensor which it cannot directly display the pressure value in unit pascal because the output from the sensor is in voltage that has been converted from resistance value.

Fig. 7. Temperature and pressure display via Blynk Apps

V. Results And Discussions

A. Testing Analysis

This system aims to provide the real-time data reading from the sensors that can be displayed on the smartphone. To achieve that goal, data communication that we used which is Bluetooth was tested to test its effectiveness. The synchronization between serial monitor in Arduino IDE and result in Blynk were tested and it shows that they were displaying the same value at the same time.

After that, the same code was tested where it can take the reading from all five sensors without any time delay. The result shows that it is impossible to read the sensor without any delay. So, to acquire the real time measurement with a short period of time delay, 0.2 second time delay has been added in the coding for each sensor to send the data to Arduino UNO.

B. Performance Analysis

In this performance analysis, there are two different time delay which are using 1 second for each sensor and another one is using 0.6 second for each sensor. Besides that, the pressure range will display from 0-100 and 90 maximum thresholds have been set for the pressure sensor. The results for pressure that above maximum threshold will be marked in red column. So, when the pressure displayed is above 90, the smartphone will vibrate and receive the warning notification.

The time delay has been set which are 1 second for each sensor to take the reading and then, 1 second additional for each round. Then, the insole was tested by standing on it for 60 seconds. Table I below shows the results for 60 seconds of standing.

TABLE I. Results For 60 Seconds Of Standing

Time (s)	Temperature (Celsius)	Pressure at Big Toe	Pressure at Meta 1	Pressure at Meta 2	Pressure at Heel
6	30.76	60	82	81	95
12	32.23	78	82	82	95
18	32.23	78	80	81	95
24	32.23	80	81	81	95
30	32.23	76	80	81	95
36	32.23	76	81	81	95
42	32.72	75	81	81	95

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48	32.72	71	81	81	95
54	32.72	54	79	80	95
60	32.23	53	79	81	95

Then, a graph pressure range versus time has been made as shown in Figure 8. It can be observed that the pressure at big toe is changing dramatically during standing. Big toe is used to maintain the body's balance while standing and it is not stay at that point [18]. That is why the pressure at big toe is not fixed during 60 seconds of standing.

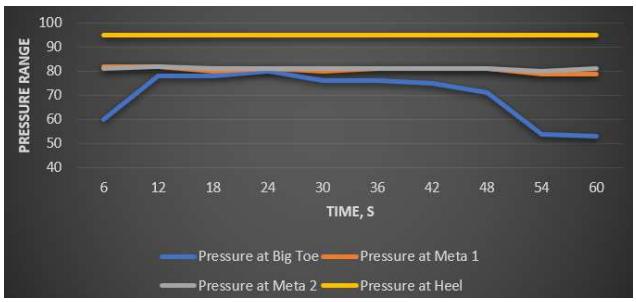

Fig. 8. Graph of pressure versus time in 60 seconds

Although the results for 1 second per sensor show the good stability in communication, but it seems not quite good in terms of effectiveness for smart insole's user due to the long data retrieval period. So, a new time delay has been set which are 0.6 seconds for all sensors. That means every 3 seconds, the sensors will take the data from the foot plantar. Table II and Figure 9 show the results recorded and graph of pressure against time for 30 seconds of standing. There were two areas of foot that show high pressure above the threshold which are heel and metatarsal 1. So, the warning message has been notified to the user as shown in Figure 10.

TABLE II. RESULTS FOR 30 SECONDS OF NORMAL STANDING

Time (seconds)	Temperature (Celsius)	Pressure at Big Toe	Pressure at Meta 1	Pressure at Meta 2	Pressure at Heel
3	20.51	69	92	88	95
6	22.46	70	92	88	95
9	20.02	68	92	88	95
12	19.04	57	92	88	95
15	20.02	64	92	87	95
18	20.51	51	92	87	95
21	21.48	52	92	86	93
24	18.07	48	92	87	96
27	21.00	40	92	87	95
30	20.51	46	92	87	95

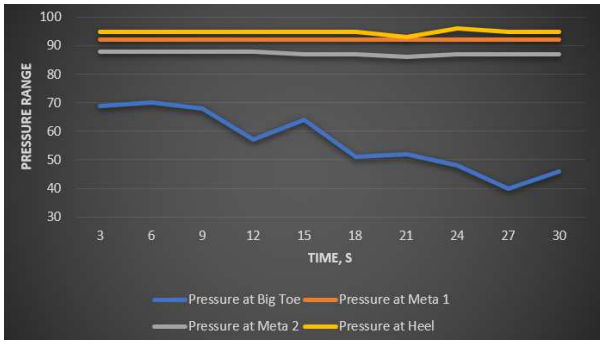

Fig. 9. Graph of pressure versus time in 30 seconds

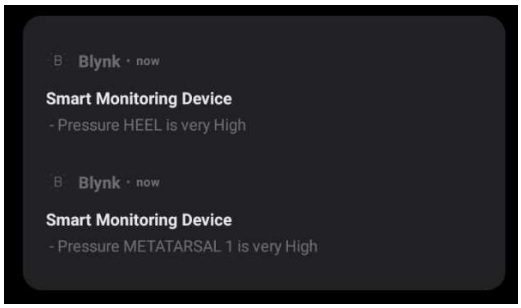

Fig. 10: Warning notification received

C. Data Analysis

A complete data set of 8 men were tested using this smart insole. This analysis aimed to compare the temperature and pressure value from 8 different people with one reading taken each person. Table III shows the result from the smart insole for the 8 different people. Besides that, Table IV provides the basic summary data for normal standing of 8 different people. The average temperature is 22.95 ˚C that is far from normal average temperature that were expected which is 30.6 ˚C but it is accepted due to the ambient temperature that can affect the foot temperature. Heel meets the expectation where it shows the highest pressure compare to others.

TABLE III. RESULTS FOR NORMAL STANDING FROM 8 DIFFERENT PEOPLE

	Temperature (Celsius)	Pressure at Big Toe	Pressure at Meta 1	Pressure at Meta 2	Pressure at Heel
Person 1	20.51	51	92	87	95
Person 2	23.44	73	79	80	96
Person 3	22.46	71	76	82	96
Person 4	28.32	79	91	88	92
Person 5	20.02	82	92	84	95
Person 6	22.46	70	92	88	95
Person 7	24.41	71	69	77	96
Person 8	21.97	68	76	76	95

TABLE IV. SUMMARY OF FOOT PRESSURE AND TEMPERATURE DATA FOR ALL 8 OF THE PARTICIPANTS

	Value Across All Participants		
	Average	Minimum	Maximum
Foot Temperature (˚C)	22.95	20.02	28.32
Pressure at Big Toe (%)	71	51	82
Pressure at Metatarsal 1 (%)	83	69	92
Pressure at Metatarsal 2 (%)	83	76	88
Pressure at Heel (%)	95	92	96

D. Walking Experiment

For this part, an experiment has been done to test whether the system is applicable for walking or not. The results for 30 seconds of normal walking have been recorded in Table V while Figure 11 illustrates the graph of pressure data against

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time. Based on Table V, the temperature result indicates the instability during walking session where the current temperature is supposed to be 21˚C, but during walking, it goes up to the highest peak of 30.76˚C and drops to the lowest peak of 12.21˚C. Wiring connection problem either loose wire or the wire is not tight may create the noise of the data reading while movement occurs especially during walking. An Arduino online forum had discussed on the instability result of the sensor that LM35 is very susceptible to interference and the circuit connection must be fully tight without using breadboard to acquire the stable result [19].

TABLE V. RESULT FOR 30 SECONDS OF WALKING

Time (seconds)	Temperature (Celsius)	Pressure at Big Toe	Pressure at Meta 1	Pressure at Meta 2	Pressure at Heel
3	21.97	47	87	78	95
6	21.00	49	85	40	93
9	30.76	0	86	36	94
12	26.85	0	88	24	36
15	23.44	0	85	77	75
18	12.21	84	23	87	83
21	21.48	85	83	32	93
24	14.16	0	88	28	0
27	20.02	0	82	91	0
30	14.65	71	88	69	39

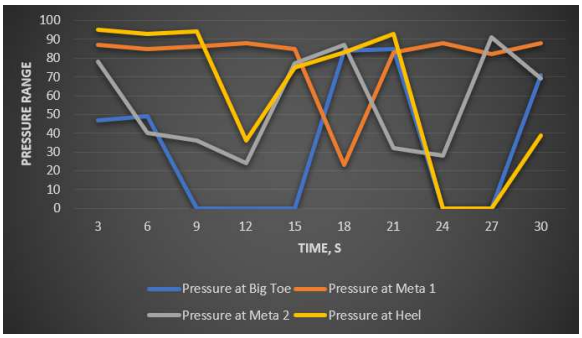

Fig. 11: Graph of pressure against time for 30 seconds of walking

Based on the graph in Figure 11 above, the plantar pressure distribution results show that the time delay used is not suitable for walking session. It is because the sensor will detect the data every 3 seconds, but each step of walking is not fix at 3 seconds. So, the smart insole device needs to be improved in real-time sensory display to ensure that the system can be used in doing various activities such as walking.

VI. CONCLUSION

This project has been successfully completed the objectives. The purpose of this project is to develop a smart insole that can limit its users in doing their daily activity to prevent from foot ulceration. This project could contribute to the medical services to monitor the diabetic patients since the number of diabetic patients are increasing every year. With a proper engineering system design, the problem can be solved by providing the diabetic patient with the best preventive measure to protect them from foot ulcer. In this work, the pressure and temperature data from foot plantar has been displayed via smartphone.

In brief, the smart insole function as an early detection device that can alert the high possibility for foot ulcer happen that influenced by the pressure and foot's temperature. It can replace the normal insole as a multifunctional insole that can be use in different type of shoes. To use the device, user needs to connect the device with a smartphone wirelessly via Bluetooth. After it succeeded, user need to open the Blynk apps that has the smart system and it will automatically display the data reading from the insole. In this project, the feedback system that can warn the user has been completed. Lastly, the system has gone through several testing and it needs some future works to ensure the stability and accuracy of the hardware used

ACKNOWLEDGEMNET

All authors have disclosed no conflicts of interest. Authors would like to thank the Ministry of Education Malaysia and Universiti Sains Malaysia. This work was supported by Universiti Sains Malaysia Short Term Grant 304/PELECT/6315342.

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